



BUGEMA

UNIVERSITY

KAMPALA CAMPUS

BSCT 3221: Open Source Software

REGNO:23/BSE/BU/R/1016

END OF Year 3, Semester 2

Introduction

The BOSC Community Library project has been developed as part of the BSCT 3221: Open Source Software Final Take-Home Examination. The project aims to demonstrate practical knowledge of open-source software development, collaborative GitHub workflows, repository governance, legal compliance, sustainability planning, and iterative software engineering practices. Throughout the development process, the system evolved from a simple static educational webpage into a responsive and interactive open-source platform featuring dynamic JSON-based resource management, category filtering, search functionality, responsive user interface design, statistics dashboards, and dark mode support.

The repository was managed using professional GitHub workflows including issue tracking, feature branching, pull requests, peer review simulation, structured commit history, and repository synchronization. Additional documentation such as contribution guidelines, legal analysis, sustainability strategies, government adoption proposals, and reflective governance discussions were also prepared to align the project with modern open-source community standards. This report presents the complete development lifecycle of the BOSC Community Library project, highlighting both the technical implementation and the professional open-source practices applied throughout the examination period.

LINKS:

<https://github.com/stevenkiwanukakafeero/BOSC-Community-Library>

BOSC Community Library : For the hosted on gitpages

My DAILY WORK PLAN

Day	Main Activity
Day 1	Repository setup + homepage
Day 2	GitHub templates + Code of Conduct
Day 3	Contributing guide + Issues
Day 4	PR Feature Enhancements #
Day 5	Refactoring , More features
Day 6	Feature Enhancement #
Bug Fix	Bug Fix #, Code review and structure arrangement
Day 7	Legal + Sustainability documents
	Reflection + Submission log + screenshots

Though at the end I have compressed the days by

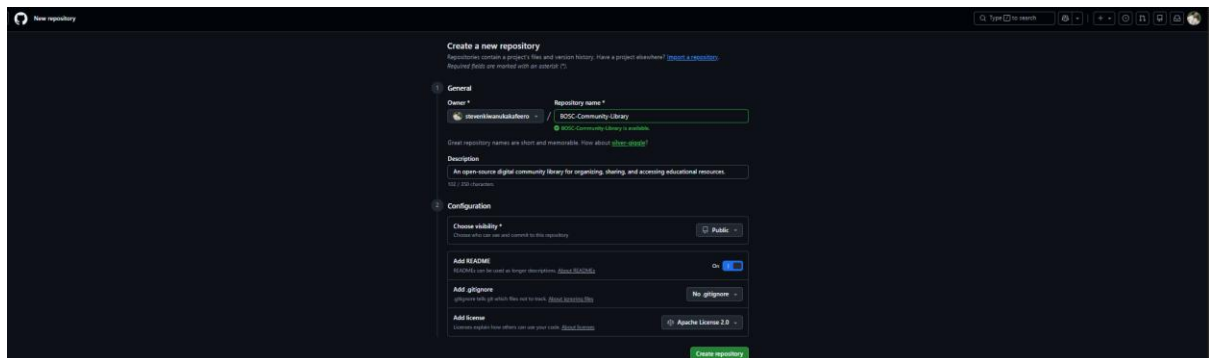
My file Structure

BOSC-Community-Library/

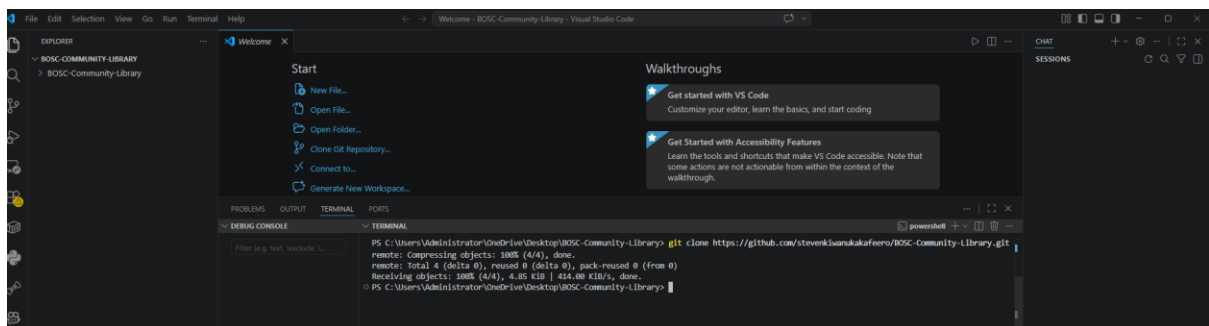
```
|
|
|— .github/
|   |— ISSUE_TEMPLATE/
|   |   └— pull_request_template.md
|
|— assets/
|   |— css/
|   |   └— js/
|
|— data/
|   └— resources.json
|
|— docs/
```

- | └─ screenshots/
- |
- | └─ reports/
- | └─ Final_Report.pdf
- |
- | └─ CODE_OF_CONDUCT.md
- | └─ CONTRIBUTING.md
- | └─ GOVERNMENT_PROPOSAL.md
- | └─ LEGAL_ANALYSIS.md
- | └─ LICENSE
- | └─ README.md
- | └─ REFLECTION.md
- | └─ SUBMISSION_LOG.md
- | └─ SUSTAINABILITY.md
- | └─ index.html
- | └─ search.html

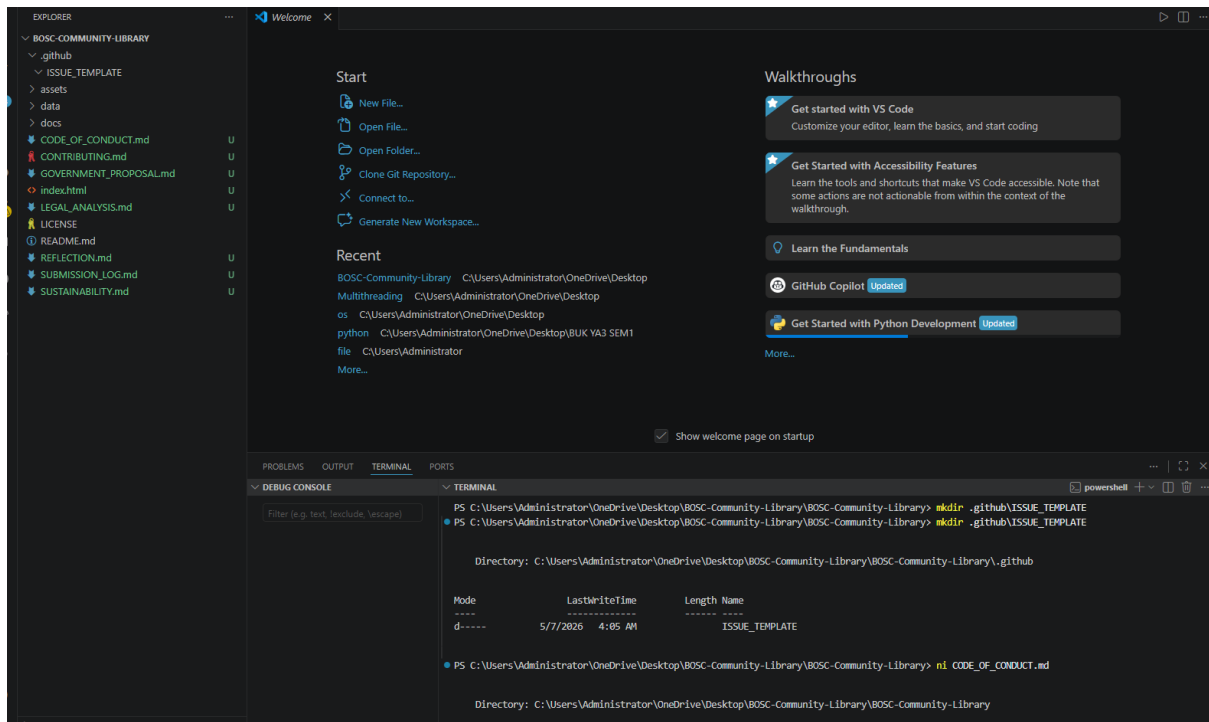
CREATING A NEW REPOSITORY



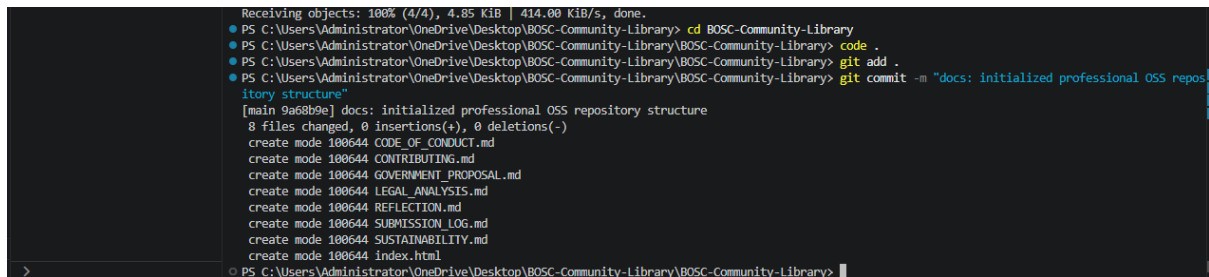
CLONING TO MY LOCAL REPOSITORY



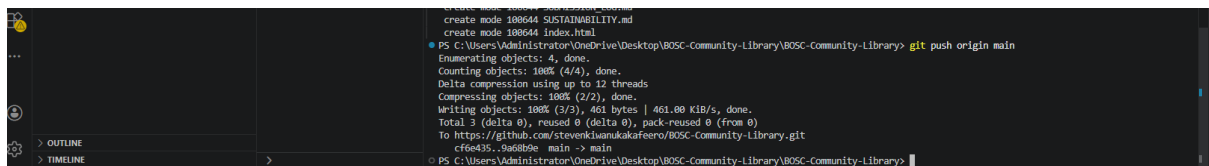
CREATING THE REQUIRED FILES



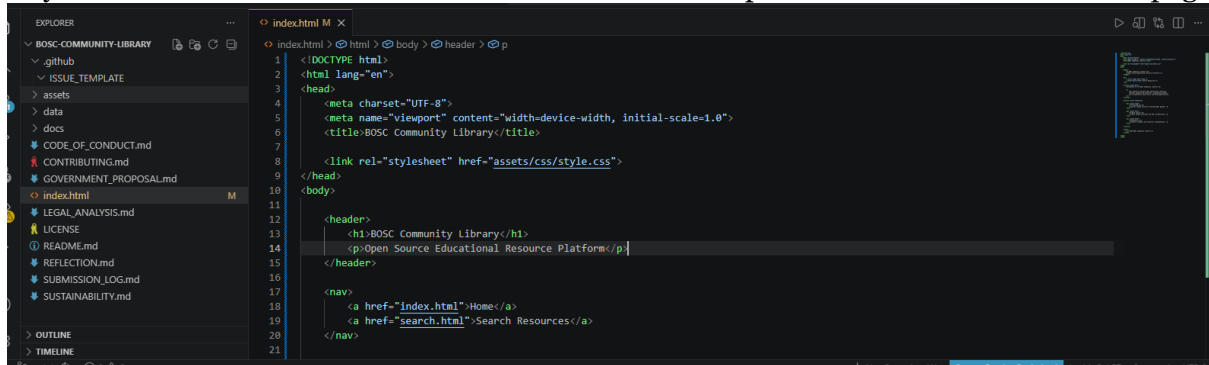
MAKING THE COMMIT



PUSHING THE CHANGES SUCCESSFULLY



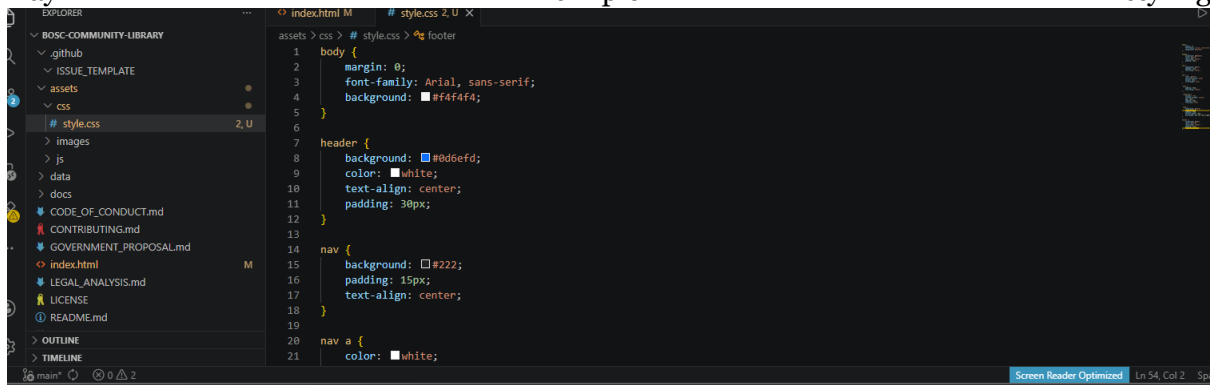
Day 1 simple webpage



Day1

simple

styling

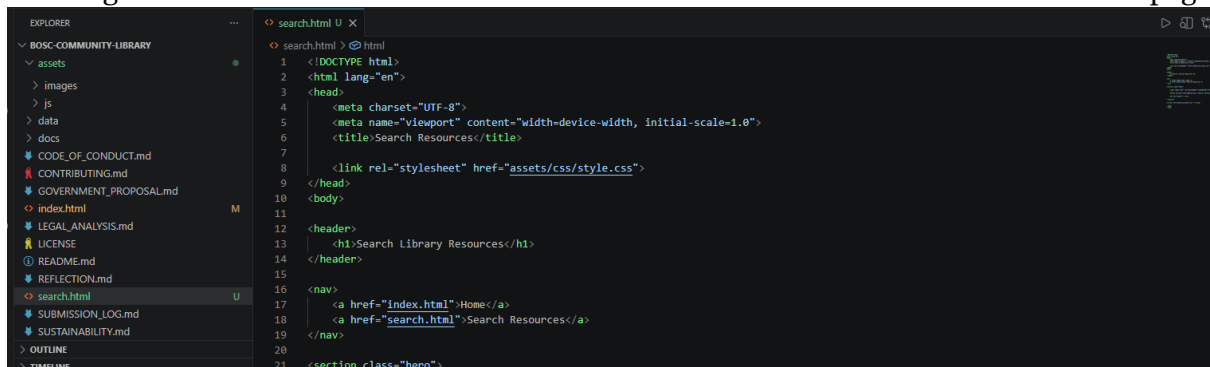


```
1 body {
2   margin: 0;
3   font-family: Arial, sans-serif;
4   background-color: #f4f4f4;
5 }
6
7 header {
8   background-color: #0d0d0d;
9   color: white;
10  text-align: center;
11  padding: 30px;
12 }
13
14 nav {
15   background-color: #222;
16   padding: 15px;
17   text-align: center;
18 }
19
20 nav a {
21   color: white;
```

Creating

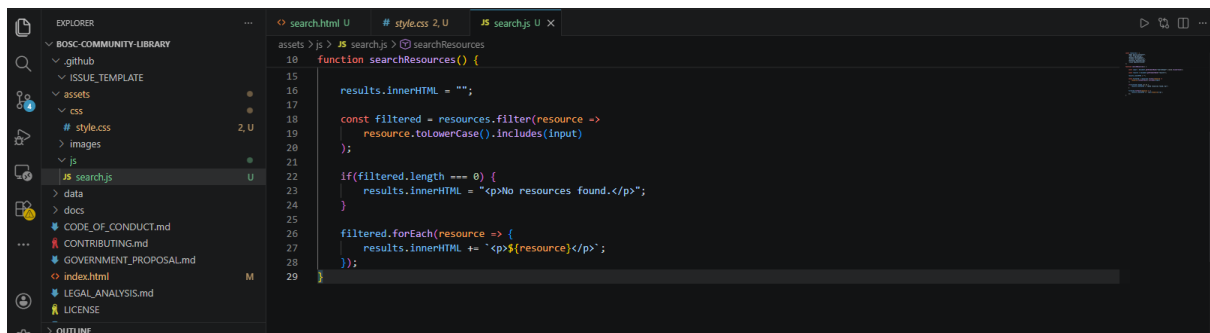
Search

page



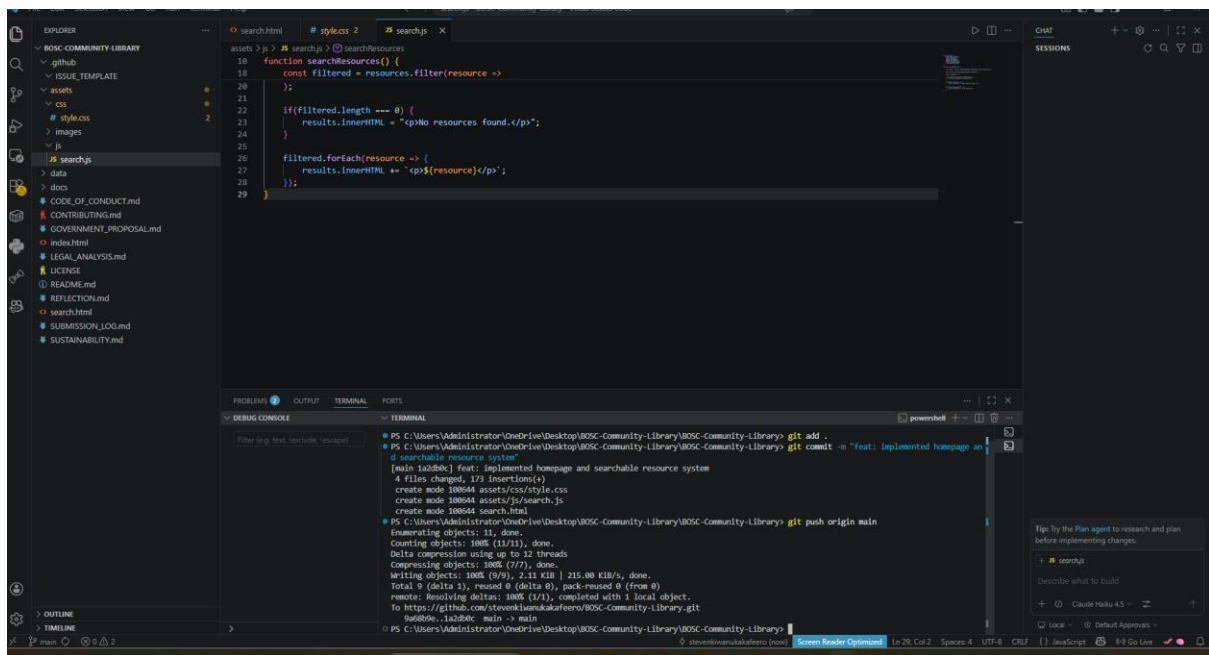
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Search Resources</title>
7   <link rel="stylesheet" href="assets/css/style.css">
8 </head>
9 <body>
10 <header>
11   <h1>Search Library Resources</h1>
12 </header>
13 <nav>
14   <a href="index.html">Home</a>
15   <a href="search.html">Search Resources</a>
16 </nav>
17 <section class="hero">
```

Advancing with Search features functionality-JavaScript

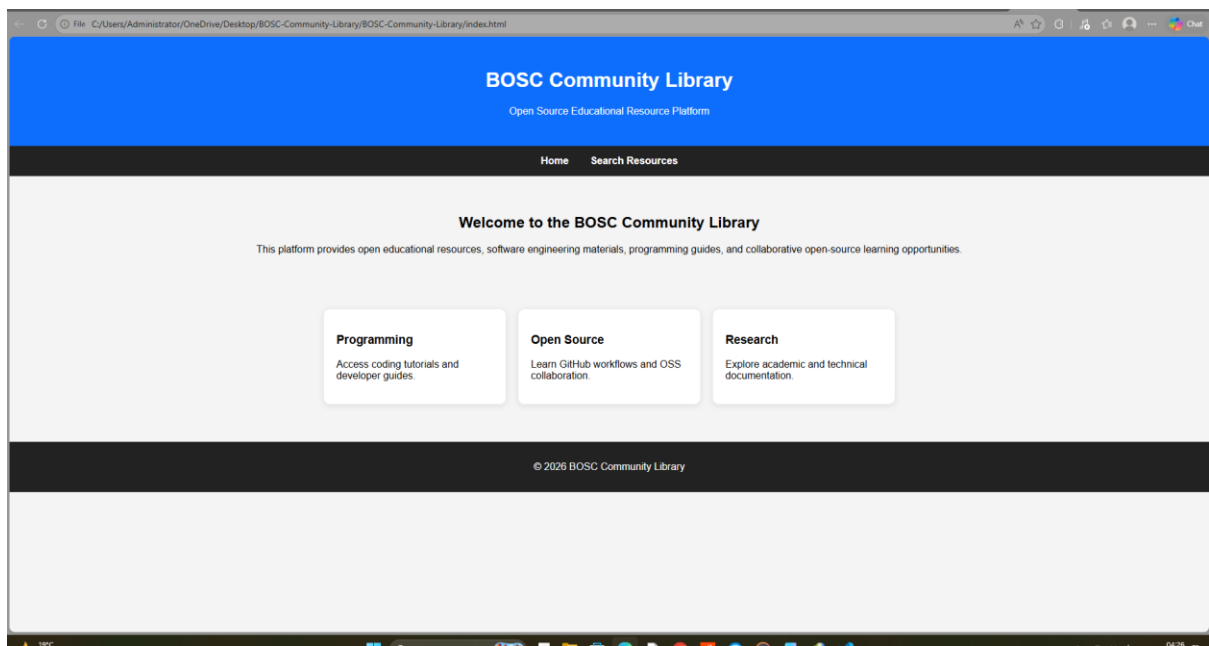


```
10 function searchResources() {
11   results.innerHTML = "";
12   const filtered = resources.filter(resource =>
13     resource.toLowerCase().includes(input)
14   );
15   if(filtered.length === 0) {
16     results.innerHTML = "<p>No resources found.</p>";
17   }
18   filtered.forEach(resource => {
19     results.innerHTML += `<p>${resource}</p>`;
20   });
21 }
```

Second Professional commit



Outcome when we open it in the browser
 .\index.html



I'm done with Day One.

DAY 2;

Today I shall:

Create				CONTRIBUTING.md
Open	your	first	GitHub	Issue
Create	a		feature	branch
Fix	a		real	issue
Commit				professionally

Push
Open
Merge PR

Pull

branch
Request

```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git pull origin main --allow-unrelated-histories
error: You have not concluded your merge (MERGE_HEAD exists).
hint: Please, commit your changes before merging.
fatal: Exiting because of unfinished merge.
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git add .
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git commit -m "Merge remote main into local main"
[main 226b363] Merge remote main into local main
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git push origin main
Enumerating objects: 5, done.
```

Started with ensuring that I pull the changes online repository to ensure that my local repository is in Syn with the local repository

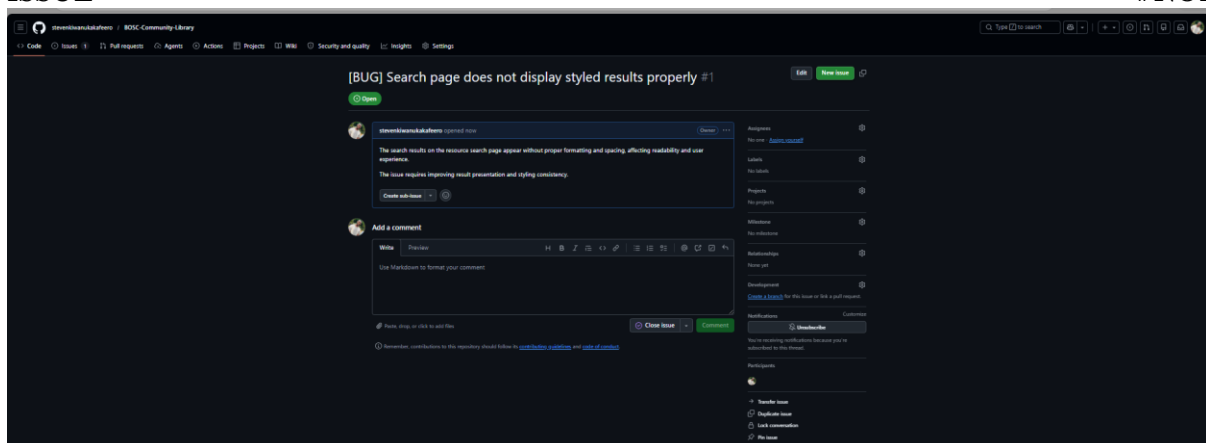
```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git status
Changes to be committed:
  new file:   CODE_OF_CONDUCT.md
  new file:   CONTRIBUTING.md
  new file:   GOVERNMENT_PROPOSAL.md
  new file:   LEGAL_ANALYSIS.md
  new file:   LICENSE
  new file:   README.md
  new file:   REFLECTION.md
  new file:   SUBMISSION_LOG.md
  new file:   SUSTAINABILITY.md
  new file:   assets/css/style.css
  new file:   assets/js/search.js
  new file:   index.html
  new file:   search.html

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
```

Made some changes

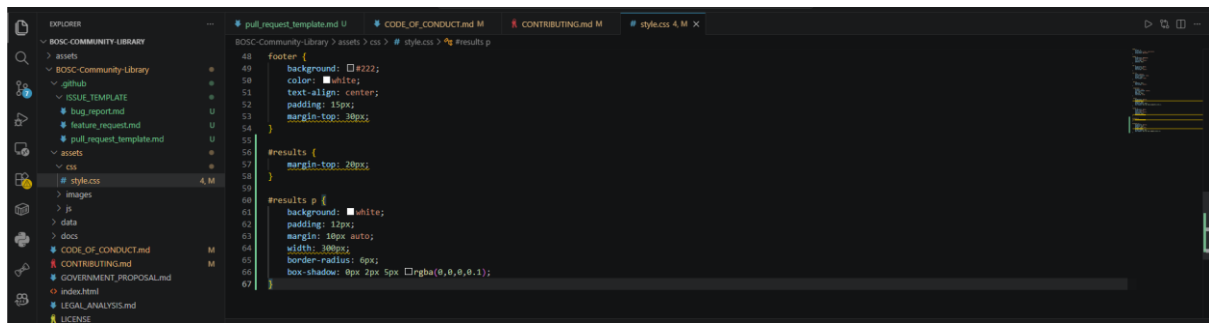
```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git commit -m "Merge remote main into local main"
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 579 bytes | 193.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/stevenkiwanukakafero/BOSC-Community-Library.git
  1a2db0c..226b363  main -> main
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git log --oneline --graph --decorate --all
* 226b363 (HEAD -> main, origin/main) Merge remote main into local main
* 1a2db0c feat: implemented homepage and searchable resource system
* 9a68b9e docs: initialized professional OSS repository structure
* cf6e435 Initial commit
* 8e4c3fc docs: More direct way and community health and collaboration templates well explained on how to add value to the Community
```

Pushed and compared
ISSUE #NO1

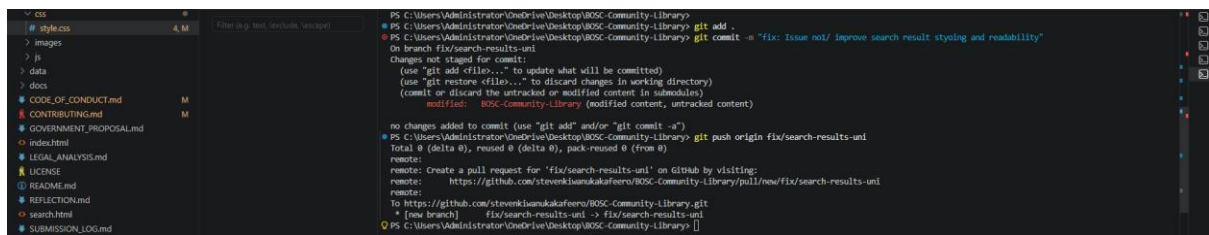


Fixing issue no1

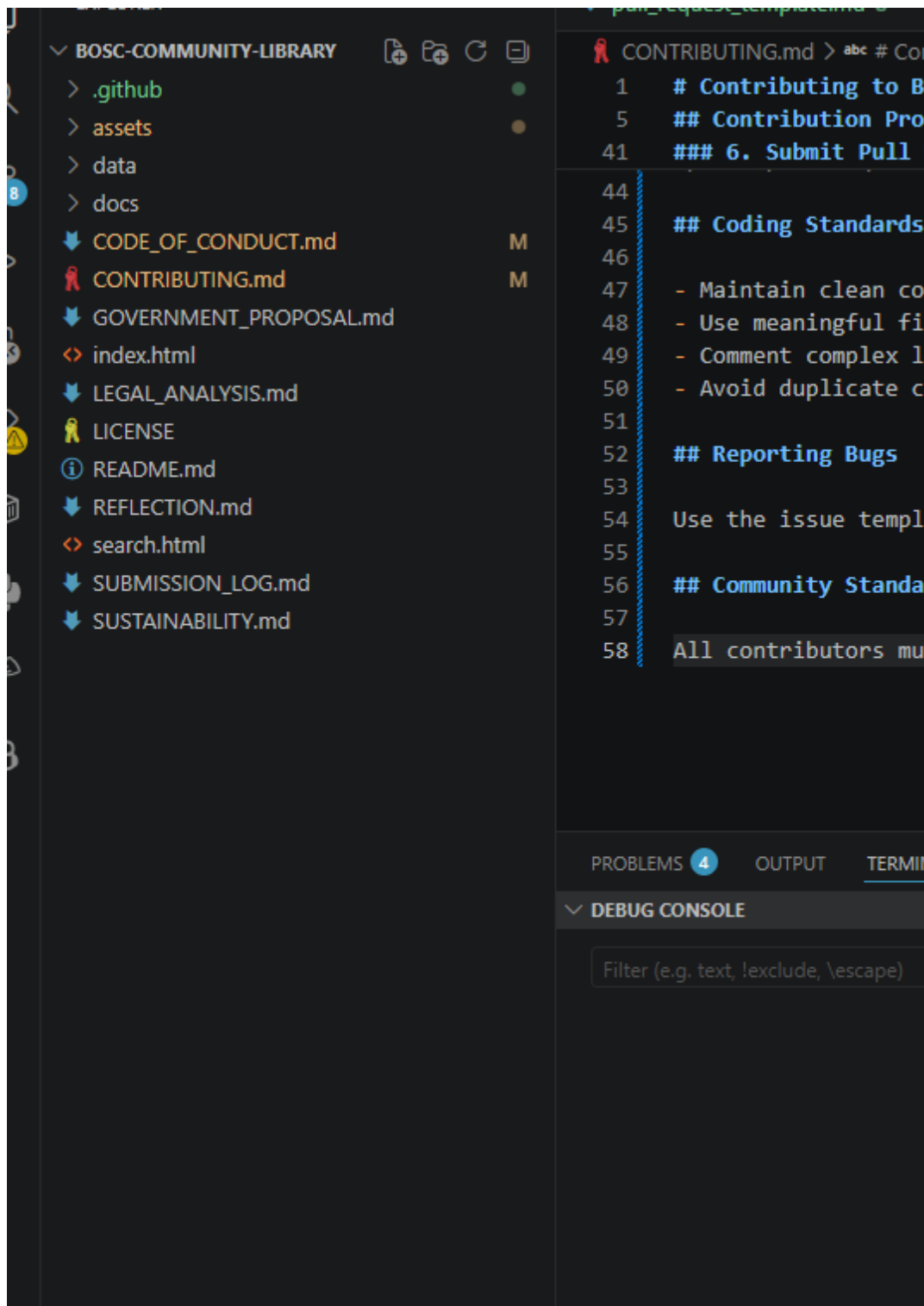
```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git pull origin main
Already up to date.
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git checkout -b fix/search-results-und
Switched to a new branch "fix/search-results-und"
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library>
```



Commit and push



this shows I have an error of multiple files; let me remove them



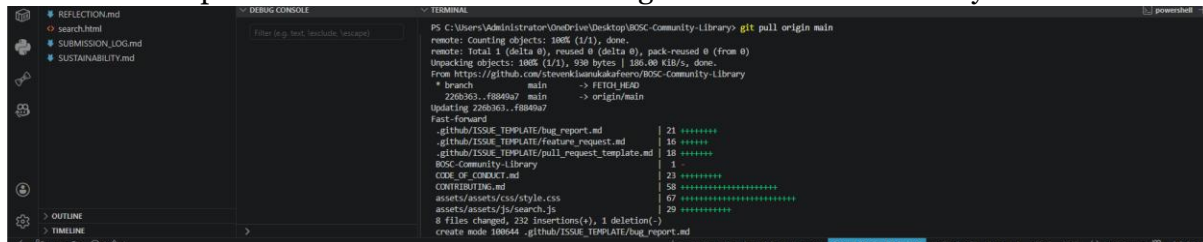
file reorganised. And I have deleted duplicates.

successfully

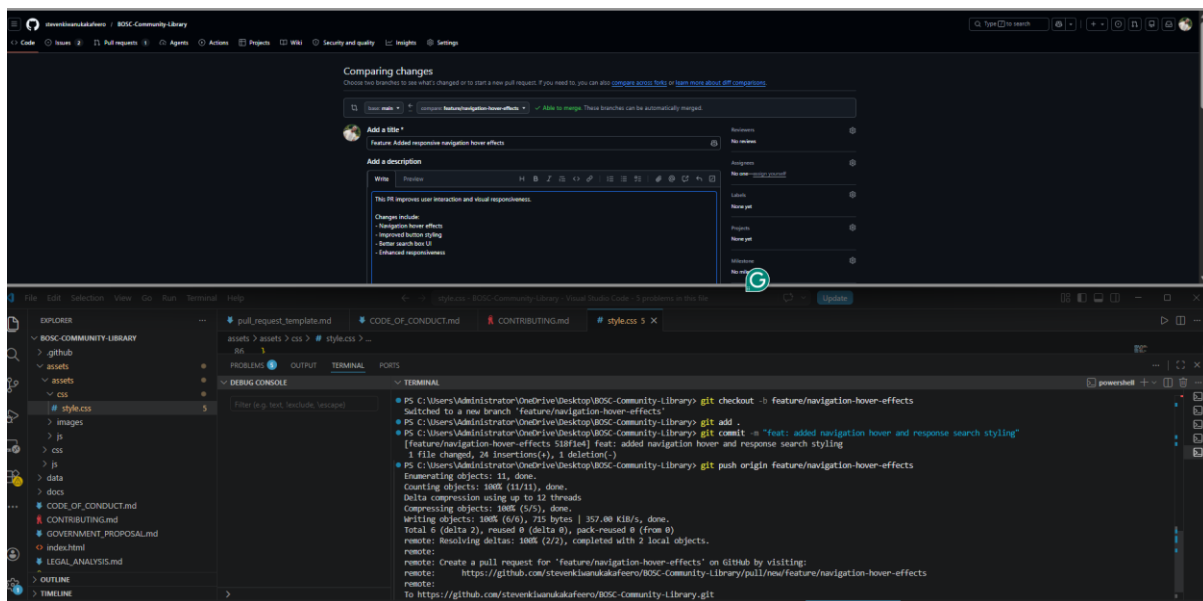
pushed

```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git push origin fix/search-result-uni
● PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git push origin fix/search-results-uni
Enumerating objects: 17, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (13/13), 2.56 KiB | 124.00 KiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/stevenkiwanukakafero/BOSC-Community-Library.git
226b363..b7b1a25  fix/search-results-uni -> fix/search-results-uni
● PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git checkout main
Switched to branch 'main'
● PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git pull origin main
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 930 bytes | 186.00 KiB/s, done.
From https://github.com/stevenkiwanukakafero/BOSC-Community-Library
```

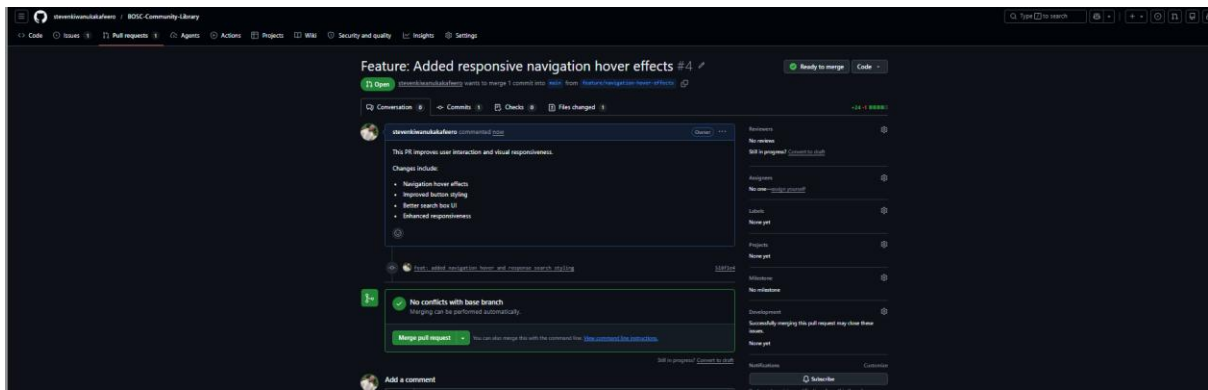
Then pulled the changes to sync them



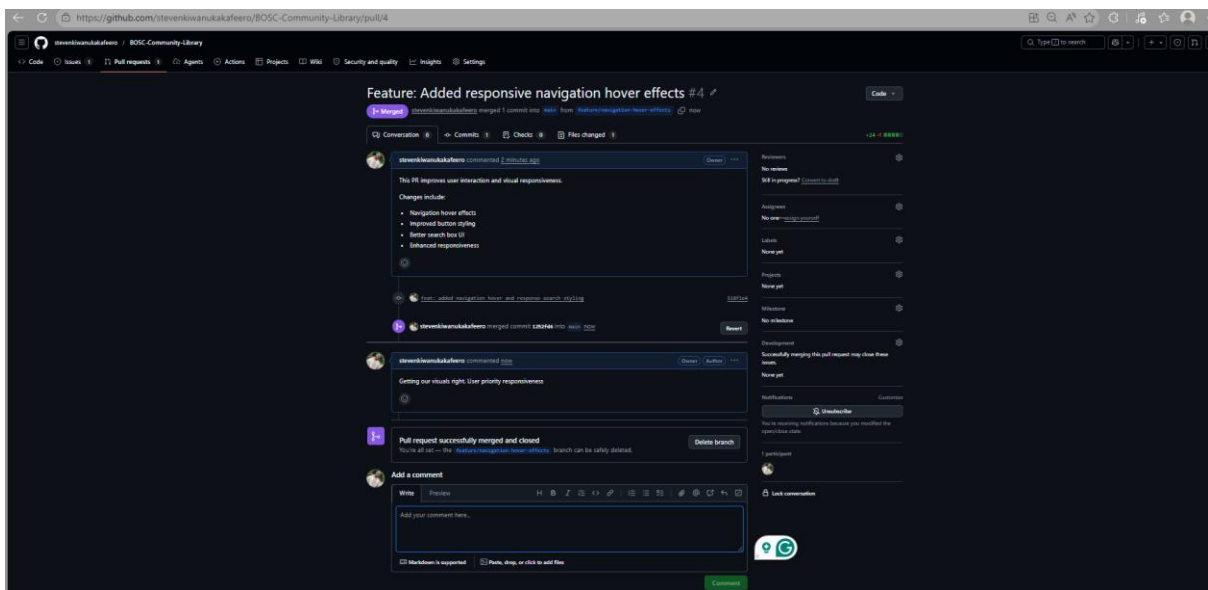
Open up issue 3: for Hover effect



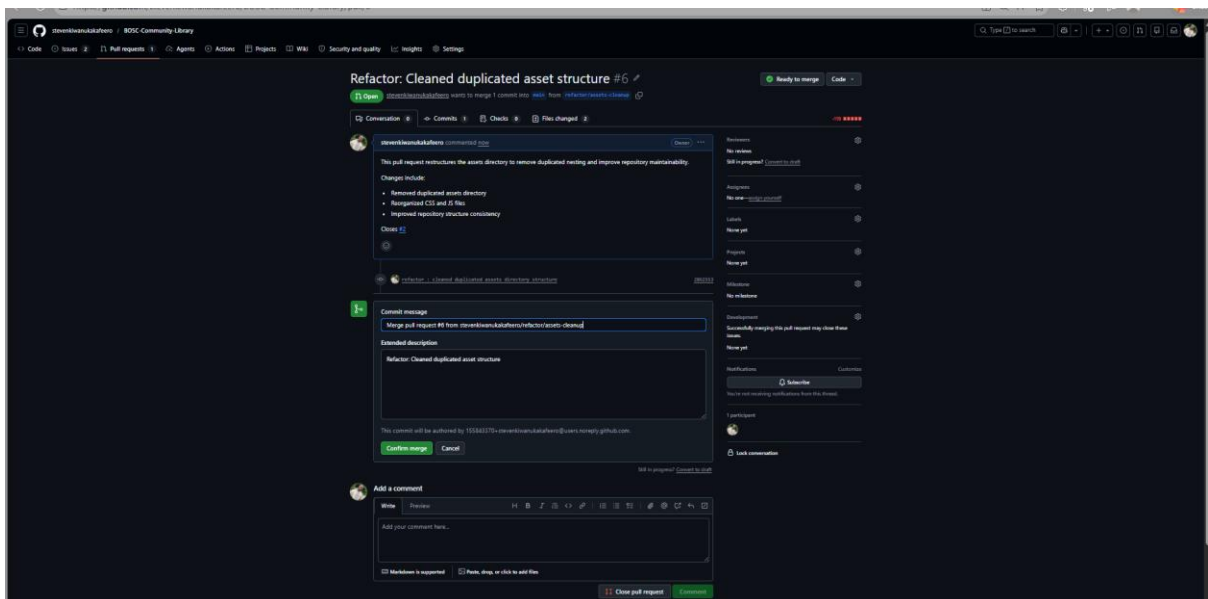
Pull request #4



Feature successfully added plus a comment



Added another refactor: Organising my file structure successfully and deleting duplicates



I have enhanced the UI and improved website interactivity and responsiveness. These included:

- Navigation hover effects
- Improved button styling
- Enhanced search form controls
- Interactive resource cards

Throughout the day, professional GitHub workflows were followed, including:

- Issue creation
- Branch management
- Commit versioning
- Pull request creation
- Peer review simulation
- Pull request merging

These activities contributed toward building a professional open-source software environment aligned with community governance and software maintenance standards.

DAY 3

Dynamic Resource Database

I shall Cover:

- stop hardcoding resources in JavaScript
- move resources into a JSON database
- dynamically load them using JavaScript fetch()

This makes my project:

- scalable
- cleaner
- more professional
- closer to real web systems

System Functionalities

- 1) JSON resource database
- 2) Dynamic loading of resources
- 3) Better search system
- 4) Dynamic resource cards
- 5) Category support
- 6) Cleaner JavaScript structure

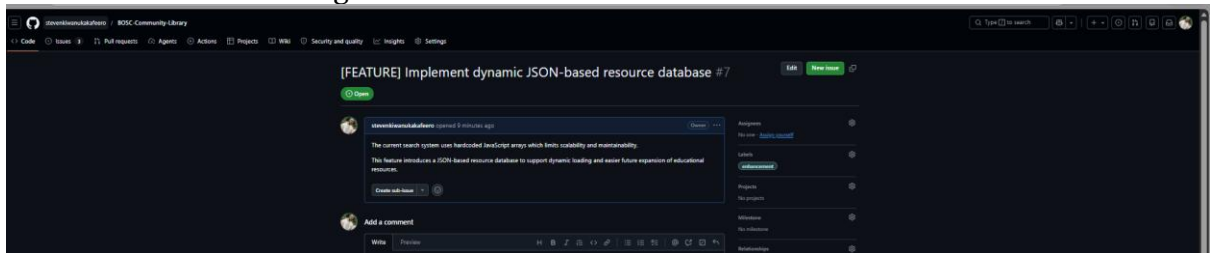
GitHub Workflow

- new issues
- new feature branches
- new pull requests
- professional commits
- peer review comments

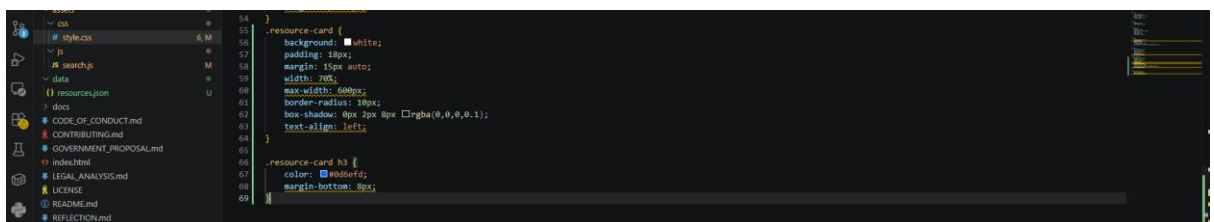
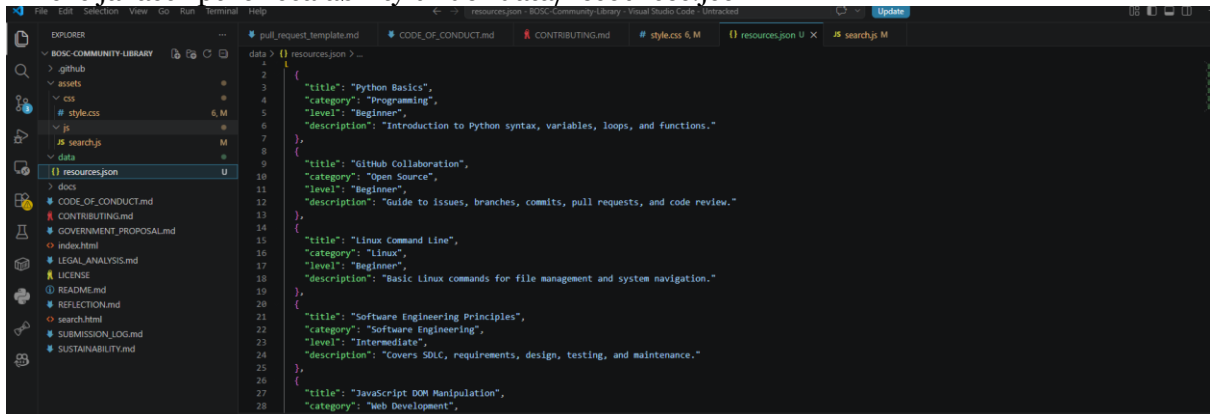
Synchronize Repositories to ensure local = online

```
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git checkout main
M
assets/css/style.css
Already on 'main'
PS C:\Users\Administrator\OneDrive\Desktop\BOSC-Community-Library> git pull origin main
From https://github.com/stevenkiwanukakafero/BOSC-Community-Library
* branch      main       -> FETCH_HEAD
Already up to date.
```

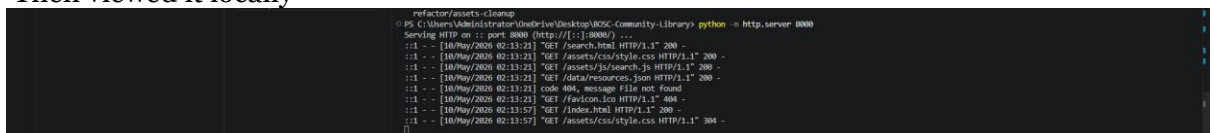
Then created an issue on github account

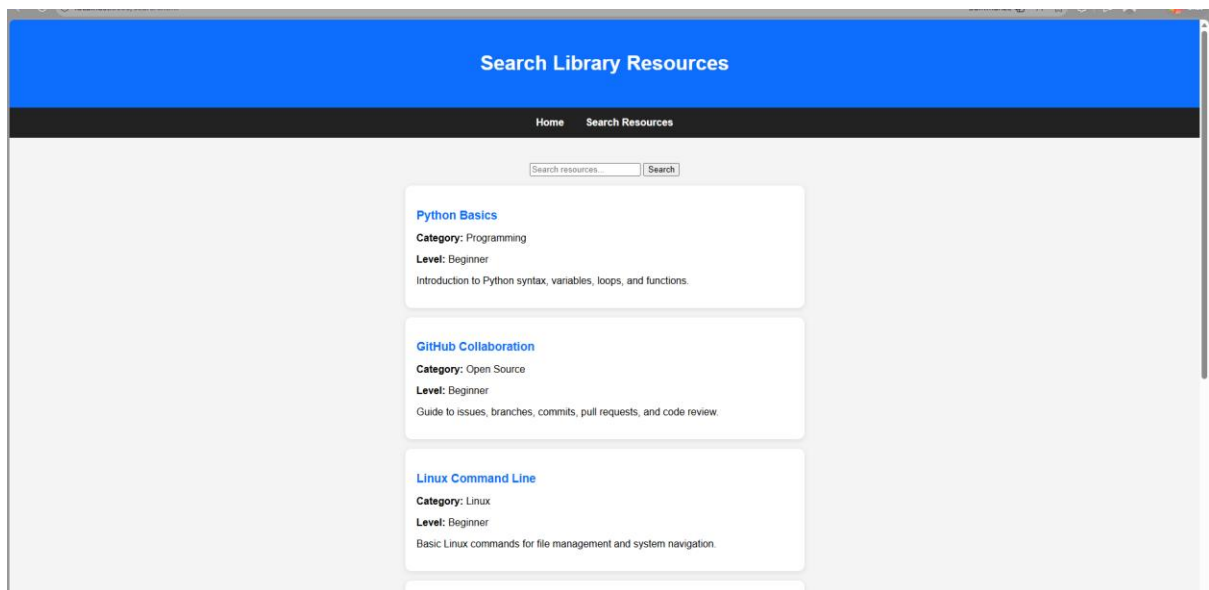


Added the cards for search bar and then styled. This is javascript for scalability under data/resources.json



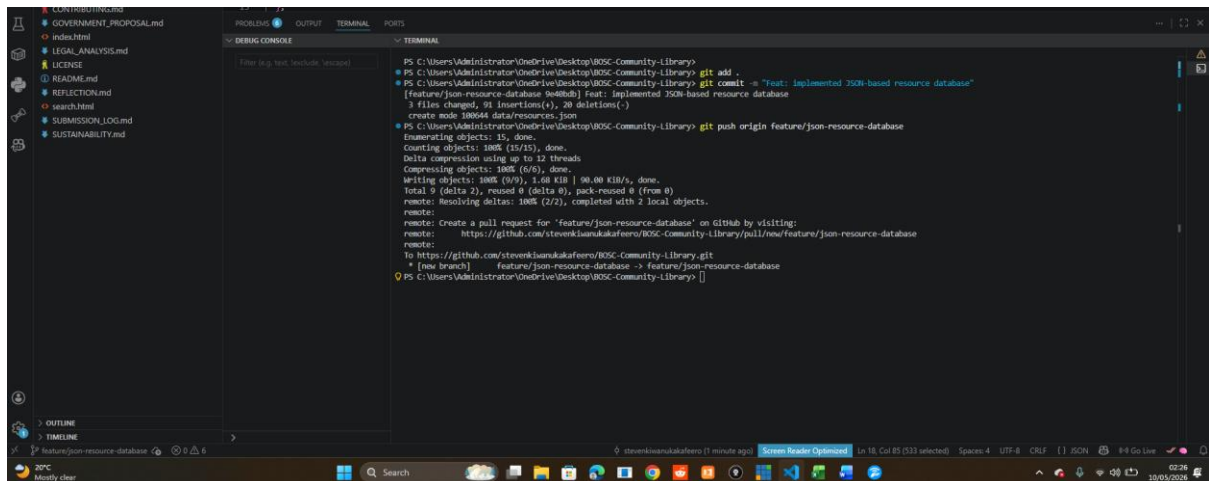
Then viewed it locally



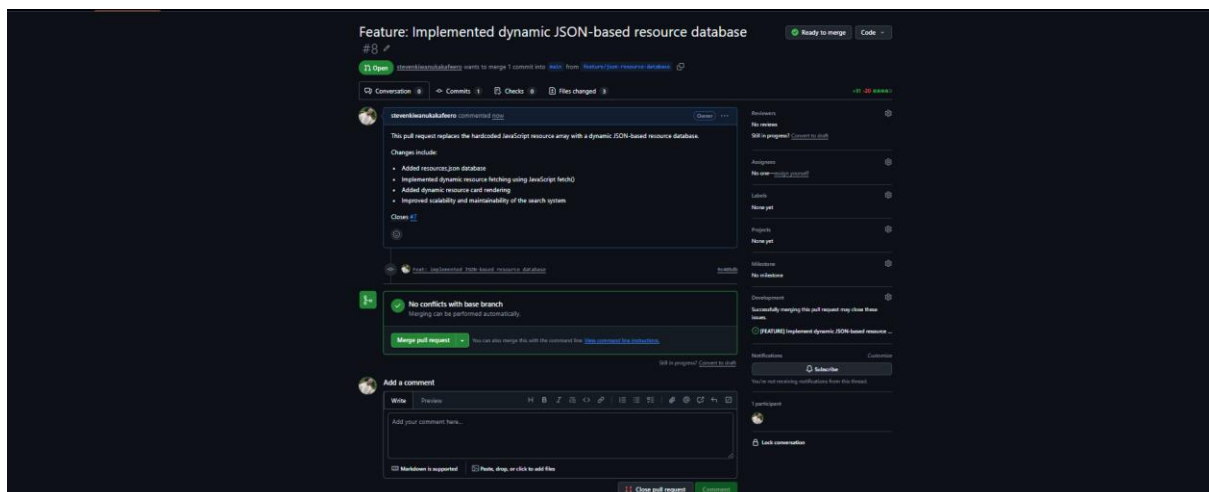


The tool search tool bar works well and verified

Then I have added and commit and pushed the changes



Then merged and closed the pull Request



This is one of the strongest technical improvements so far because my project now demonstrates:

Feature	Technology
Dynamic data loading	JavaScript fetch()
Structured data storage	JSON
Dynamic rendering	DOM manipulation
Search filtering	JavaScript
OSS workflow	GitHub PR workflow

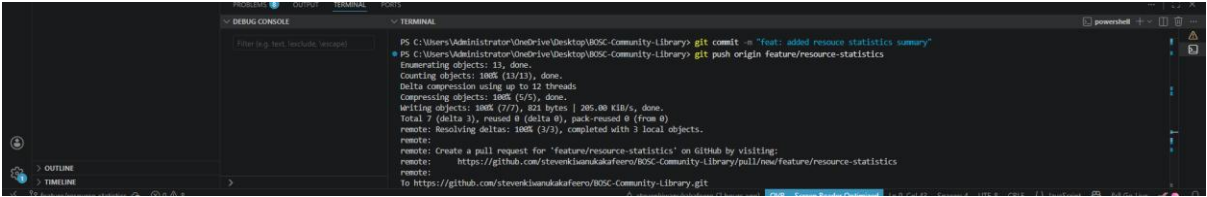
New issue



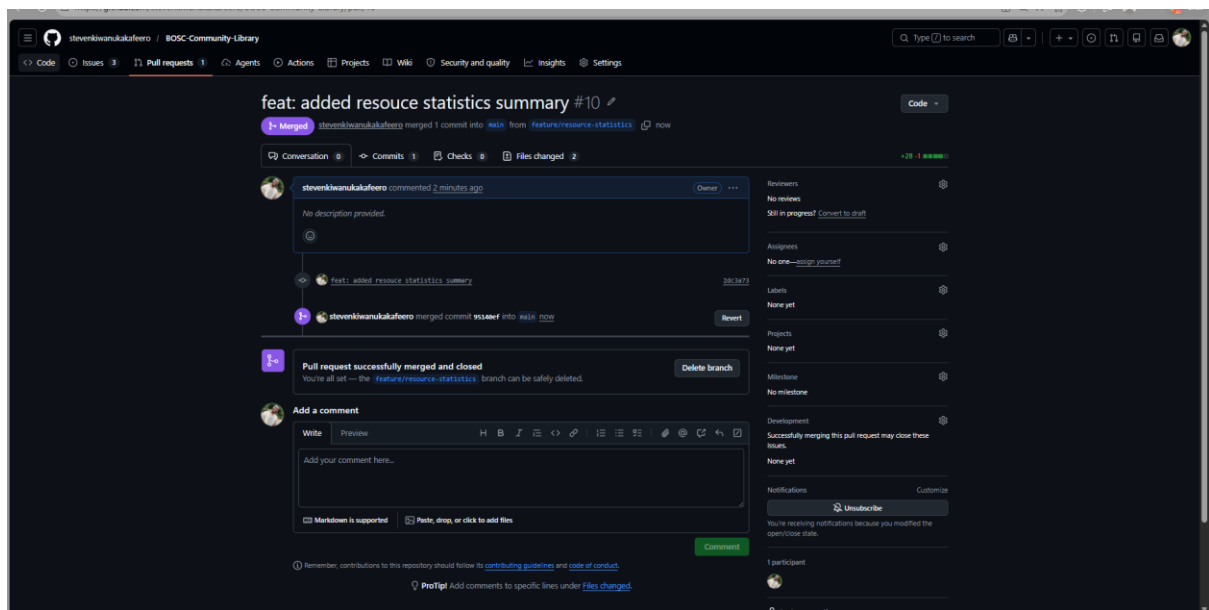
code

```
55 document.addEventListener("DOMContentLoaded", loadResources);
56
57 function displayStats() {
58   const stats = document.getElementById("stats");
59   const categories = [...new Set(resources.map(resource => resource.category))];
60
61   stats.innerHTML = `
62     <p><strong>Total Resources:</strong> ${resources.length}</p>
63     <p><strong>Categories:</strong> ${categories.length}</p>
64   `;
65 }
```

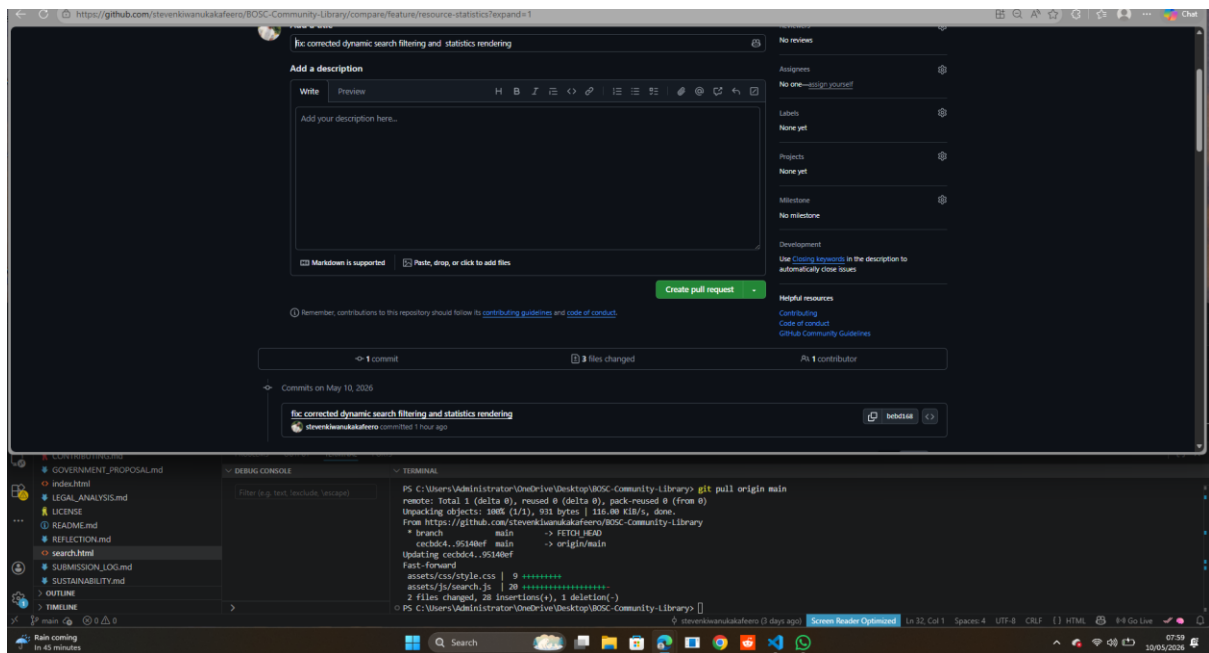
Committed and pushed the changes



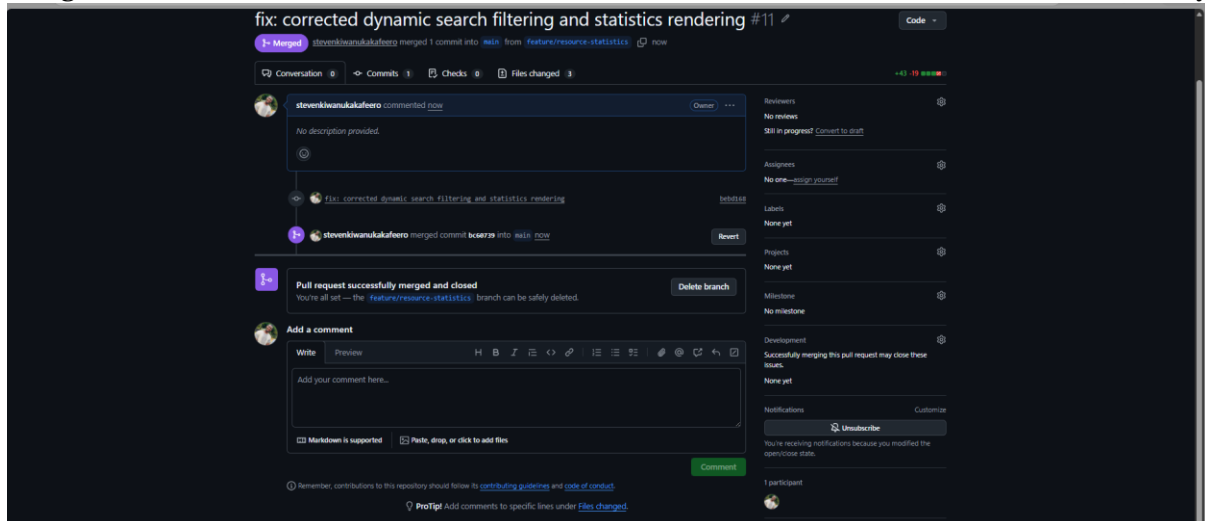
Then I have merged day



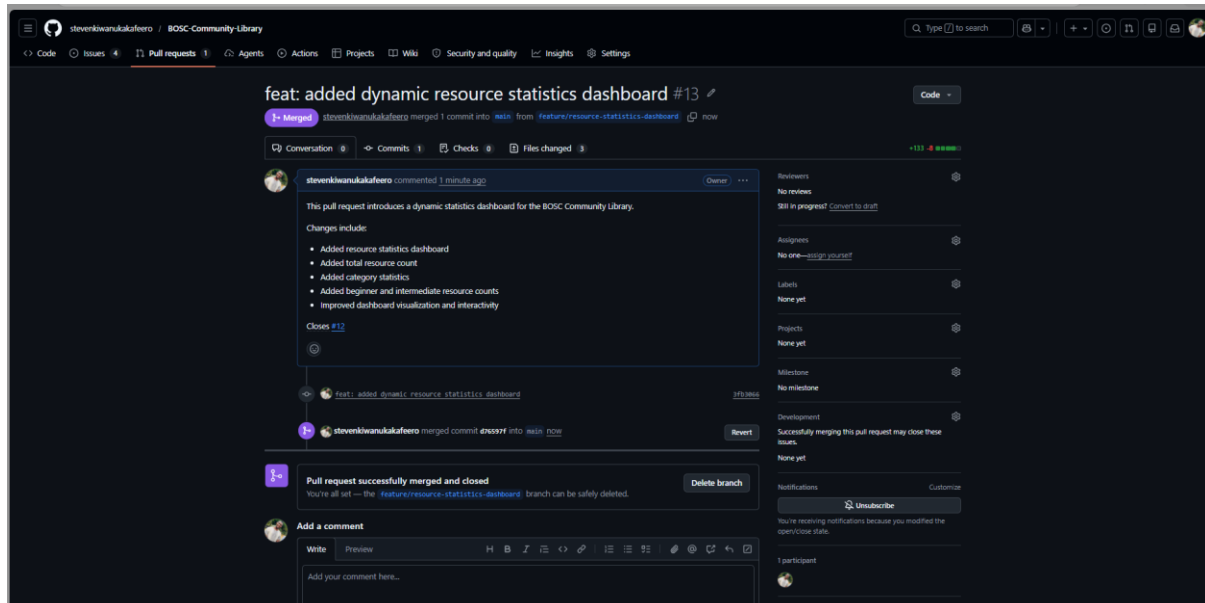
Creating another dynamic search filtering and statistics rendering



Merged and closed successfully



Added a dashboard feature



WHAT I HAVE ACHIEVED IN DAY 3

Core Technical Features

Dynamic	JSON	resource	database
JavaScript		fetch()	implementation
Dynamic		resource	rendering
Search			functionality
Category			filtering
Dynamic	statistics		dashboard
Resource			cards
Improved UI structure			

GitHub / OSS Workflow

Multiple	issues	created
Multiple	feature	branches
Professional	commit	history
Pull		requests
Peer	review	comments
Merged		PRs
Debugging and maintenance workflow		

DAY 4

UI/UX Professionalization & Advanced Frontend Features

Now I shall implement:

- ✓ the dynamic system works
- ✓ JSON database works
- ✓ filtering works
- ✓ dashboard works

Day 4, I will focus on making the platform:

- 🚀 modern
- 🚀 responsive
- 🚀 visually impressive
- 🚀 more interactive

1. Homepage Redesign

Then I improve:

- hero section
- cards
- layout
- project presentation

Goal:

make the homepage look like a real educational platform.

2. Dark Mode Toggle

Users can switch:

- Light mode
- Dark mode

This is a very modern frontend feature.

3. Responsive Mobile Layout

I optimize:

- dashboard
- cards
- navigation
- search controls

for smaller screens.

4. Dashboard Improvement

Then I enhance:

- stat card appearance
- spacing
- colors
- hover effects

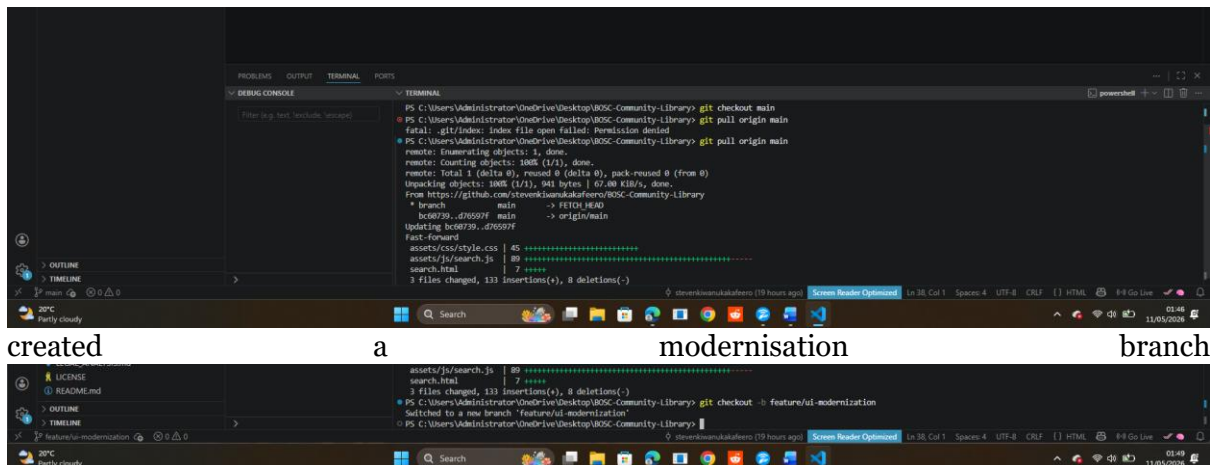
5. Then I shall add:

- smooth hover transitions
- card effects
- subtle animations

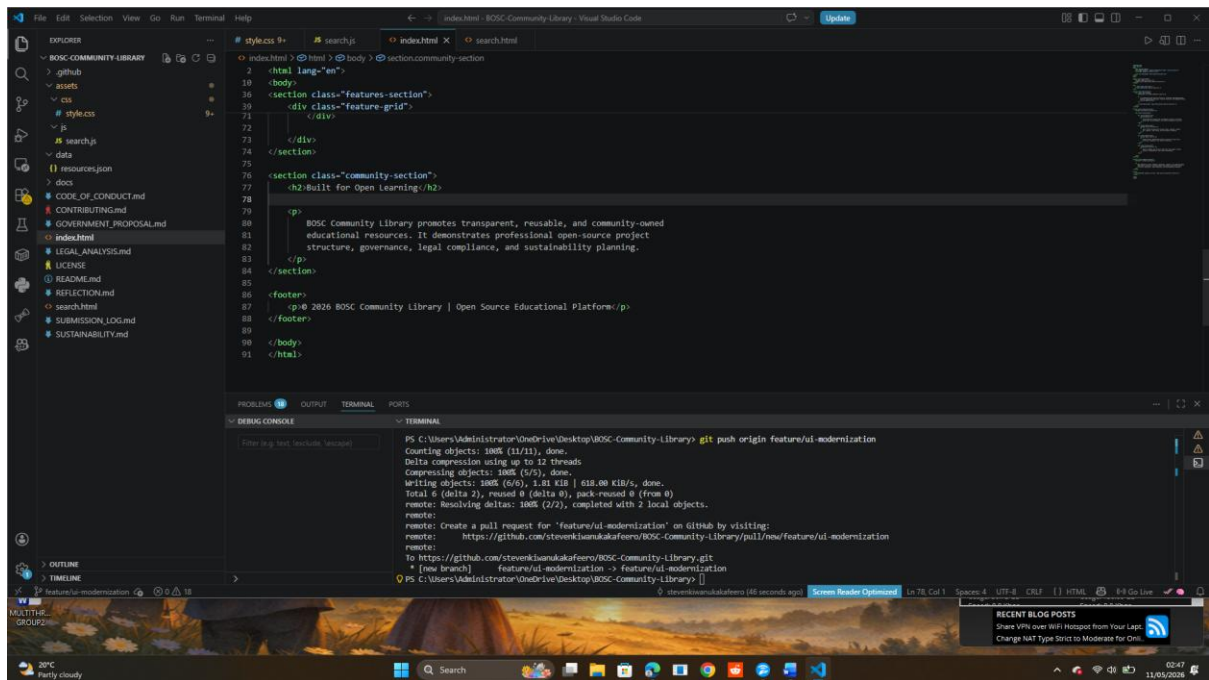
Better

Animations

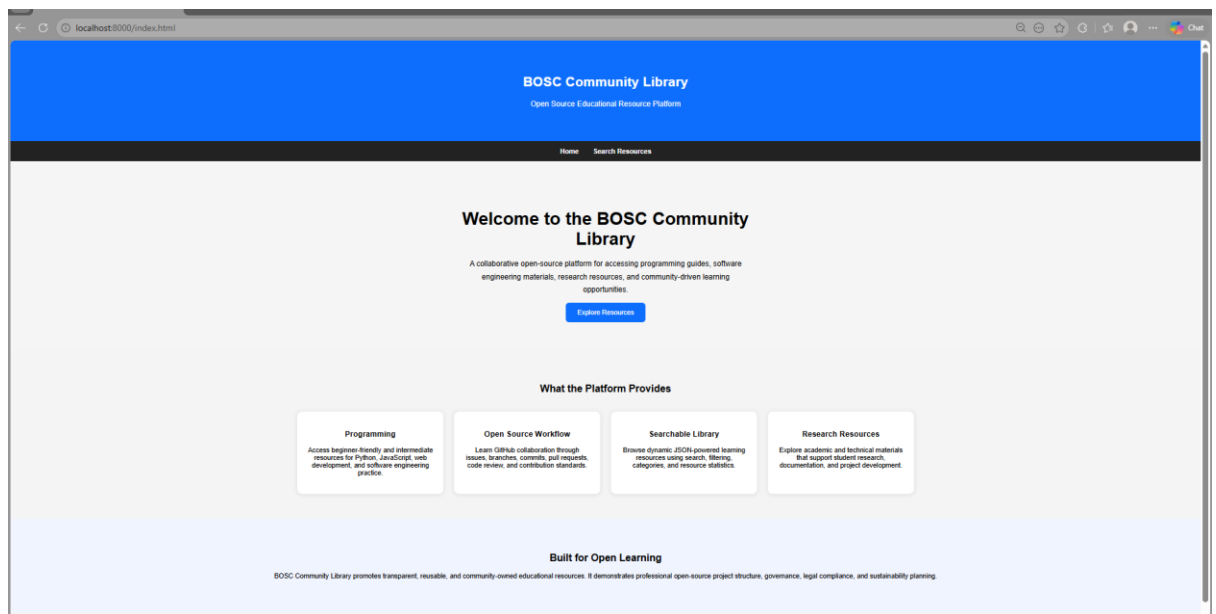
Synchronised the repositories local = online



Improving the index.html and css

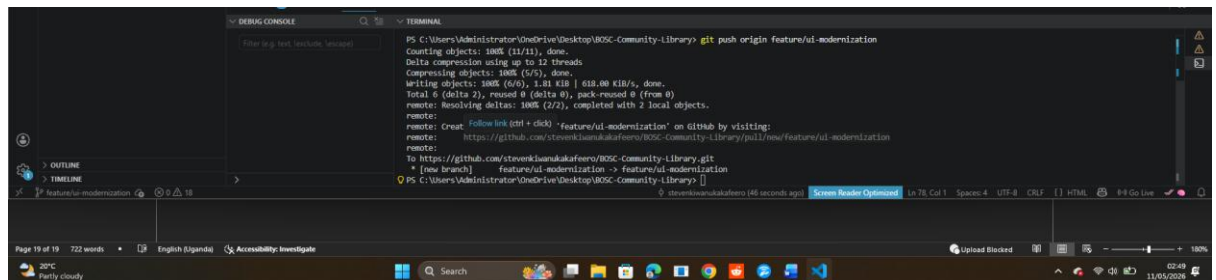


When viewed on the local host:



Git add.

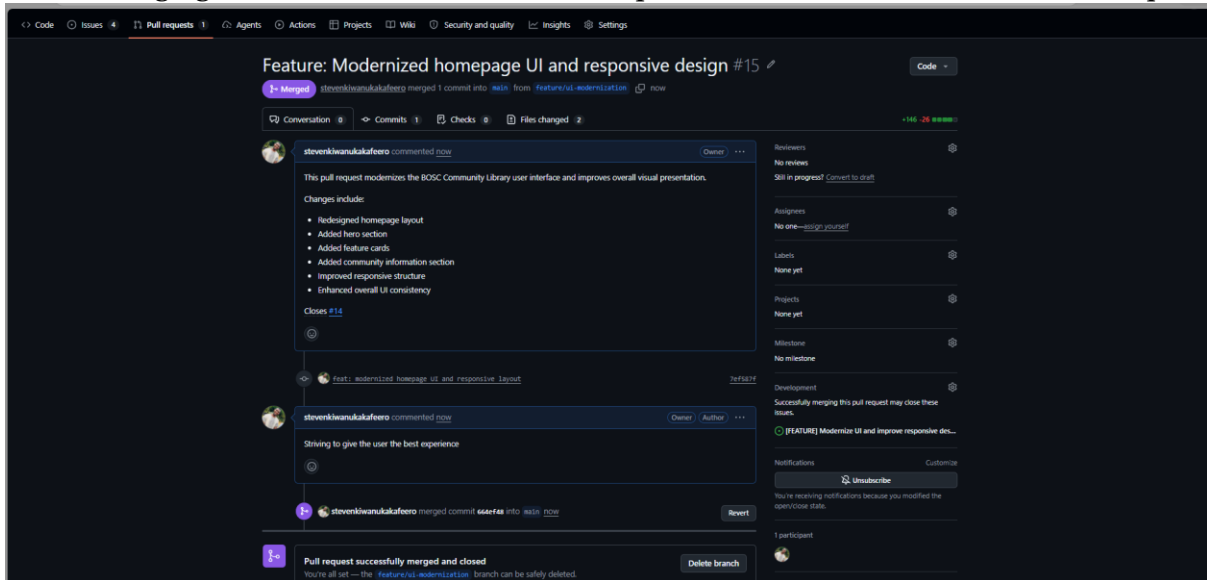
Git push the changes and merging



Merging

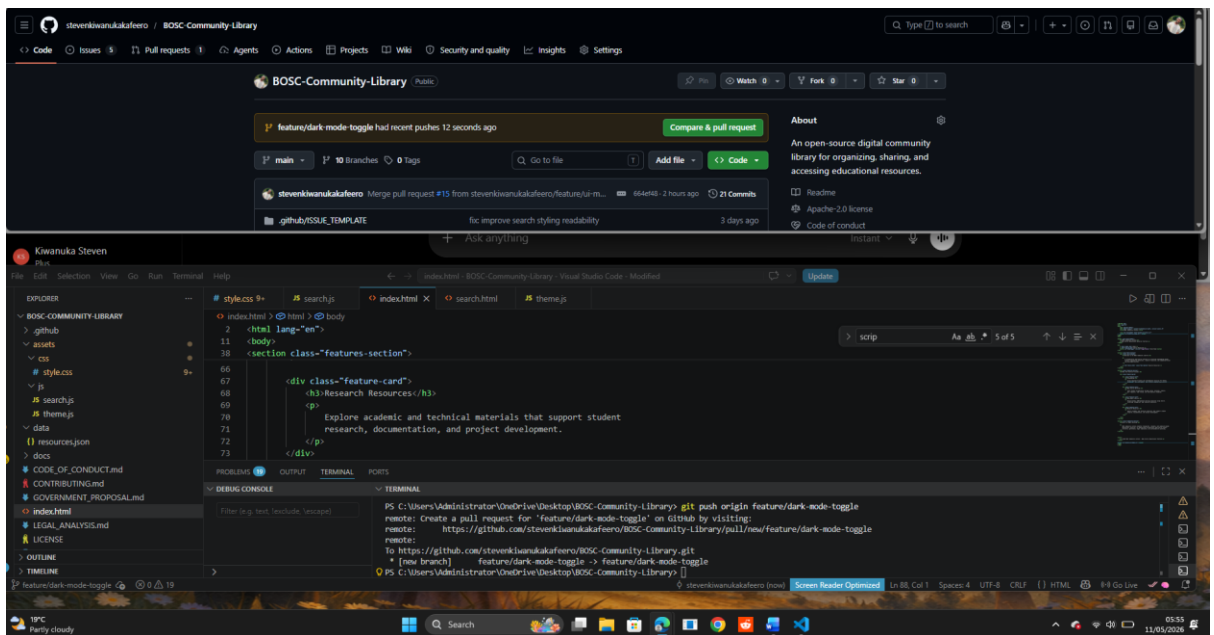
pull

request



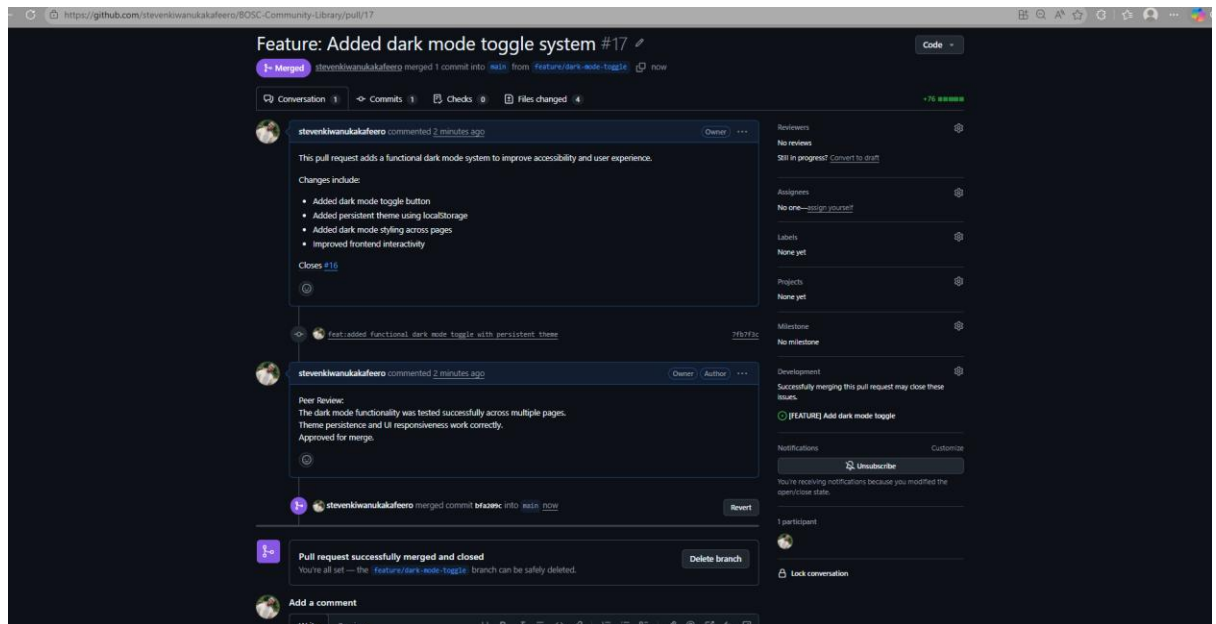
Schronise the local repository again after pushing changes

Another feature for dark mode toogle

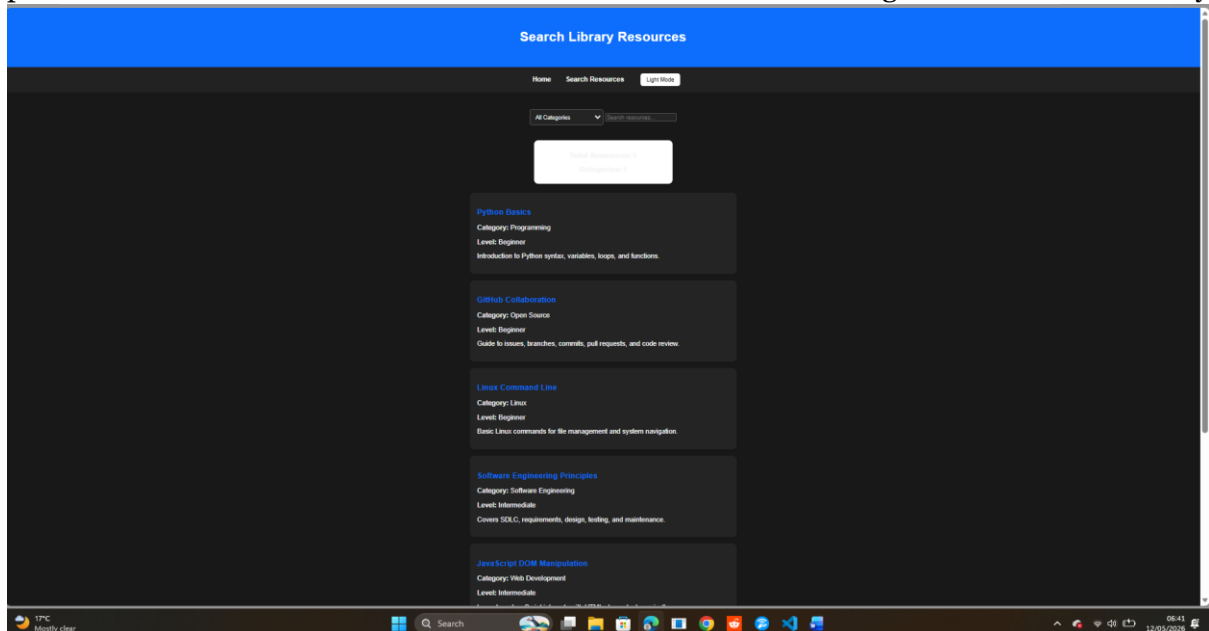


Finished working with Dark mode

Now merging

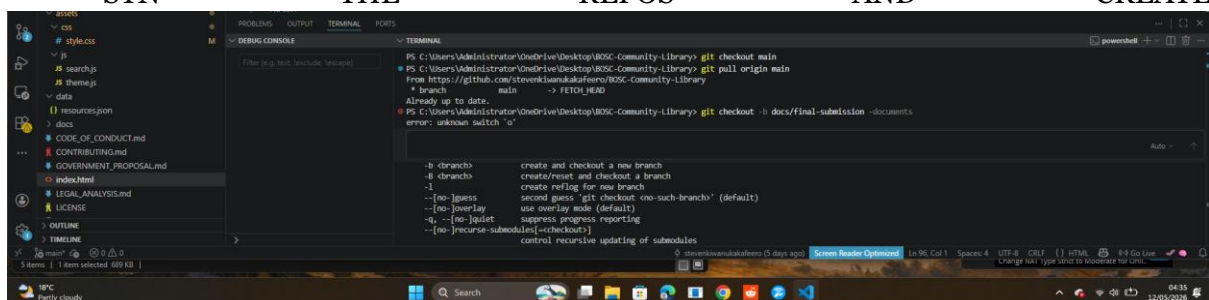


peer comment added and merged successfully

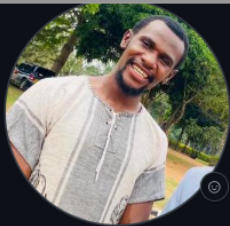


LAST DAY SUBMISSION

SYN THE REPOS AND CREATE



MY WEEKLY REVIE:



stevenkiwanukakafeero

Edit profile

3 followers · 6 following

Achievements



Organizations



You unlocked new Achievements with private contributions! Show them off by including private contributions in your Profile in settings.

stevenkiwanukakafeero / README.md

Hi! I'm STEVEN KIWANUKA KAFEERO - Final year student at Bugema University as 2026. Pursuing A Bachelor of Sciences In Software Engineering

- I'm interested in CODING, ROBOTICS,ARTIFICIAL INTELLIGENCE AND POLITICS.
- I'm currently learning MASTERING JAVA
- I'm looking to collaborate WITH OPEN ELEMENTS , OPENMRS, OSS Companies, SYMETRIX AV company AND OTHER COMPANIES THAT MIGHT IGINITE THE GIAT IN ME AND TO BUILDIND A MORE WHOLESUM IT WORLD.
- How to reach me ...+256779668768 OR kiwanukasteven21@gmail.com
- Pronouns: he/ him.
- Fun fact: ...The Lie" said to "the Truth", "Let's take a bath together, the well water is very nice". "The Truth", still suspicious, tested the water and found out it really was nice
- So they got naked and bathed. But suddenly, the Lie leapt out of the water and fled, wearing the clothes of the Truth.

The Truth, furious, climbed out of the well to get her clothes back. But the World, upon seeing the naked Truth, looked away, with anger and contempt. Poor Truth returned to the well and disappeared forever, hiding her shame.

Since then, the Lie runs around the world, dressed as the Truth, and society is very happy ... because the world has no desire to know the naked Truth."

Pinned

OpenELUS-Global-2

Forked from DIGI-LIW/OpenELUS-Global-2

This is the core repo for OpenELUS global v3

Java 1

BOS

Forked from bos-com/BOS

BOS (Bugema Operating System) the internet OSI Free, Open Source, Self hostable

1

GPA-CAL

TO CALCULATE THE GPA OF THE STUDENTS PERFORMANCE

GreenCode

Forked from bos-com/GreenCode

GreenCode: Empowering sustainable agriculture through ICT, innovation, and digital solutions for smarter learning farming

Java 1

LifeLine-ICT

Forked from bos-com/LifeLine-ICT

Welcome to the ICT-Driven Disaster Preparedness & Early Warning project by the Bugema Open Source Community (BOSC). We build open-source tools to protect communities from climate risks like floods ...

Python 1

lost_and_found

HTML

152 contributions in the last year

May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May

Mon

Wed

Fri

Learn how we count contributions

Less More

I have earned 3 badge achievements,



stevenkiwanukakafeero

Edit profile

3 followers · 6 following

Achievements



Organizations



Earn +



Pull Shark x2

@stevenkiwanukakafeero opened pull requests that have been merged.

History

- Bronze unlocked · First unlocked on Apr 30
- stevenkiwanukakafeero/OpenCare-Core#1 · 2nd pull request merged
- stevenkiwanukakafeero/BOSC-Community-Library#6 · 16th pull request merged

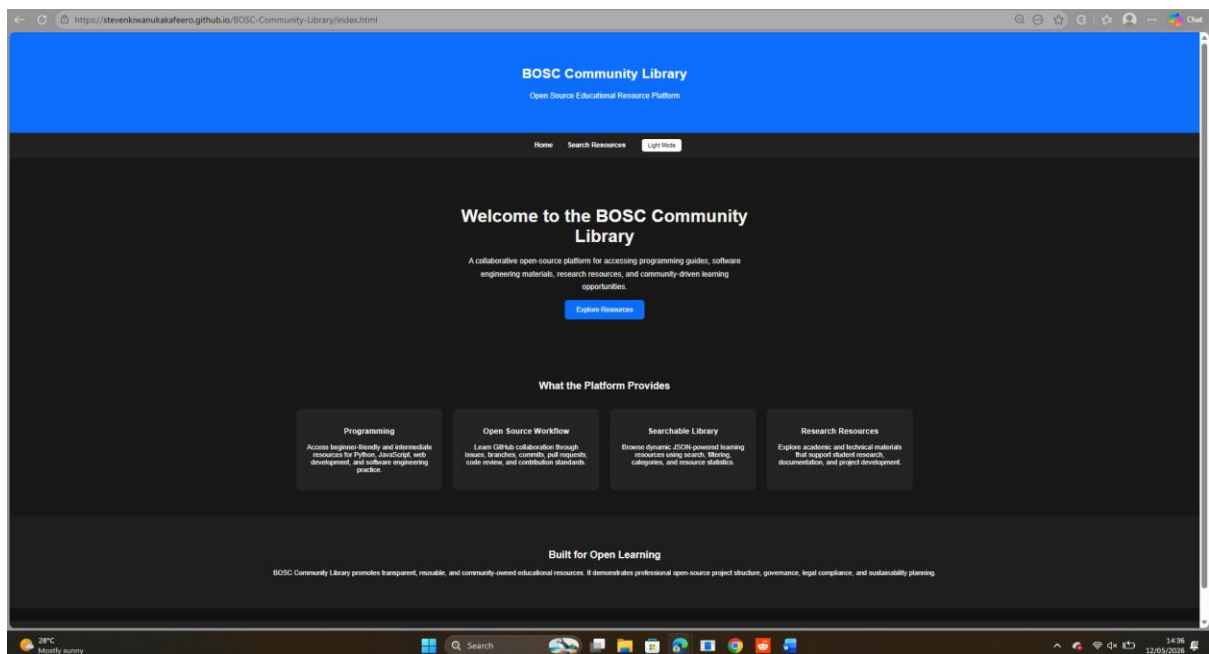


Quickdraw

and a *2 for pull shark after my core java concepts being merged in OpenCare and Bosc

Finally, I have hosted it on gitpages

[BOSC Community Library](https://stevenkwanakafeero.github.io/BOSC-Community-Library/index.html)



Conclusion

The BOSC Community Library project successfully fulfilled the objectives of the BSCT 3221 Open Source Software take-home examination by combining technical development, professional GitHub workflows, and open-source governance practices. Throughout the project, the system evolved from a simple static webpage into a responsive and interactive educational platform featuring dynamic JSON resource management, search and filtering functionality, statistics dashboards, responsive layouts, and dark mode support. The project also demonstrated practical use of issue tracking, branching strategies, pull requests, peer review simulation, repository maintenance, and documentation standards. Overall, the experience strengthened my understanding of collaborative software development and sustainable open-source project management.